

Synthetic Publicity: Common Knowledge and Heteronomous Bayesian Updating in Generative Information Environments

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Abstract

Generative systems can alter public belief without persuading each individual that a proposition is true. They can instead alter what individuals think other people believe, how independent those apparent beliefs are, and whether a signal appears publicly recognized. This paper calls that condition **synthetic publicity**: perceived public visibility or social corroboration unsupported by an equivalent population of independent human speakers, witnesses, or endorsers. The concept joins two bodies of theory that are usually treated separately. Common-knowledge theory explains how public signals and higher-order beliefs enable coordination. Heteronomous Bayesian Updating explains how an agent assigns epistemic weight to another actor's reaction, especially when that reaction is mistaken for an independent judgment. Their interaction matters in generative environments because one process can produce many apparently distinct reactions, while recommender systems and visible metrics can make those reactions appear mutually observed. The paper offers a bounded conceptual model, distinguishes synthetic publicity from ordinary repetition, social proof, astroturfing, and strict common knowledge, and derives falsifiable predictions. It does not claim that synthetic content presently dominates political behavior or that artificial signals invariably deceive users. Its contribution is an analytical vocabulary and an experimental agenda for testing when public visibility becomes detached from human independence.

Keywords: common knowledge; higher-order beliefs; Bayesian updating; generative artificial intelligence; public signals; perceived consensus; social bots; coordination

1. Introduction

A person scrolling through a feed sees hundreds of messages endorsing the same claim. She may remain unsure whether the claim is true. Yet she can become confident that other people believe it, that they know others believe it, and that opposing it carries a social cost. Her factual belief has moved little. Her expectation of collective behavior has moved a great deal.

This difference is easy to lose when online influence is described only as persuasion or misinformation. A message can change a person's estimate of the world. It can also change her estimate of other people's estimates. The second movement can affect coordination, silence, mobilization, compliance, or enforcement even when the first is weak.

Steven Pinker's synthesis of common-knowledge research provides one half of the mechanism. Common knowledge is not merely information held by many people. It includes knowledge that others know, knowledge that others know that others know, and so on. In practice, conspicuous public events allow people to treat this otherwise impossible recursion as settled. Pinker also makes a narrower observation about digital media: personalized feeds do not literally recreate the nationwide common knowledge once generated by a small number of broadcast networks, but virality can create a sense of common knowledge among the people who matter to a user.

The other half comes from Heteronomous Bayesian Updating (HBU), a concept developed in the

author’s prior work on legal metaphors and AI sycophancy. HBU describes a person who calibrates a belief through an interlocutor’s apparent reaction rather than through direct observation of outcomes. This is not necessarily irrational. Other people often possess relevant information. The failure arises when a reaction optimized for engagement, compliance, reassurance, or replication is treated as an independent assessment of truth.

Generative systems connect these levels. They can make one process look like many speakers, make dependent outputs look like independent corroboration, and make selected visibility look like population-level endorsement. This paper calls the resulting condition synthetic publicity.

The claim is conceptual and conditional. It is not that every generated message is deceptive, every viral trend is artificial, or every user mistakes popularity for truth. Nor does the paper claim priority over research on astroturfing, computational propaganda, artificial consensus, correlated social learning, or higher-order beliefs. Its narrower contribution is to integrate three mechanisms within one testable information-design construct: dependent signals are multiplied or selectively displayed; that display represents them as social plurality; and an observer infers that the display is visible to, or recognized by, other observers.

2. Three epistemic levels

Let (H) be a proposition about the world, (s) an observed signal, and (a_i) the action of agent (i) . At the first level, an agent updates a factual belief:

$$P_i(H | s) = \frac{P_i(s | H)P_i(H)}{P_i(s)}.$$

At the second level, the agent infers a social distribution:

$$P_i(B_j(H) > \tau | s),$$

where $B_j(H)$ is agent j ’s degree of belief in H , and τ is a confidence threshold.

At the third level, the agent estimates whether the social estimate is itself shared:

$$P_i(P_j(B_k(H) > \tau) > \rho | s).$$

The notation need not continue indefinitely in a human mind. Common-knowledge theory explains why a public or self-evident event can stand in for the recursive hierarchy. When everyone can see the event and see that others can see it, coordination may become rational. Monderer and Samet’s common p -belief relaxes certainty: sufficiently confident higher-order belief can support coordination when its level exceeds a threshold determined by the costs of miscoordination.

These levels can move separately. A visible trend can have little effect on $P_i(H)$ while sharply changing $P_i(B_j(H) > \tau)$. If the payoff from action depends on what others will do, the second change can dominate the first.

Four states must remain distinct. Individual exposure means only that agent i observes s . A public signal is observed by a defined group under a distribution rule that is itself sufficiently known. Perceived publicness means that i believes others have observed or recognized s , whether or not

that belief is correct. Strict common knowledge adds the indefinitely iterated epistemic condition required by formal results. Personalized feeds ordinarily establish, at most, perceived publicness. Strict common knowledge remains a benchmark, not an empirical default.

3. Heteronomous Bayesian Updating

Bayesian learning normally treats signals according to their expected reliability and dependence. In social life, however, the signal often consists of another actor’s reaction. A judge notices how colleagues receive an argument. A regulator observes which description experts repeat. A user asks a conversational system whether an intuition is reasonable. The reaction becomes evidence.

Call the received reaction r_j . A stylized update is:

$$\log O_i(H \mid r_j) = \log O_i(H) + w_{ij} \log LR(r_j),$$

where O_i denotes agent i ’s odds and LR the likelihood ratio attributed to the semantic content of the reaction. The separate term w_{ij} represents the social evidentiary multiplier the observer assigns to its source:

$$w_{ij}^{perceived} = f(A_j, C_j, Ind_j, Aut_j, Pub_j),$$

where A_j is attributed authority, C_j expressed confidence, Ind_j perceived independence from other displayed sources, Aut_j attributed autonomy of judgment, and Pub_j perceived public recognition. To avoid duplicating reliability already encoded in LR , HBU refers specifically to updating in which these attributed social properties supply incremental evidentiary weight. Its empirical content lies in measuring those attributions and testing their predictive value over a standard correlated-signal model.

HBU becomes pathological when perceived weight exceeds warranted weight:

$$w_{ij}^{perceived} > w_{ij}^{warranted}.$$

Generative systems create several routes to this inequality. A system may be optimized to maintain engagement or agreement. Multiple outputs may share a model, prompt lineage, retrieval source, or optimization target. A platform may select the outputs most likely to provoke further reaction. The user can nevertheless experience each output as another spontaneous judgment.

The problem is not simply that a model might be wrong. It is that the social meaning of its reaction can be misclassified. An optimized response is read as an autonomous witness; repeated descendants are read as independent sources; visibility selected by an interface is read as prevalence in a population.

4. From public signals to synthetic publicity

Strict common knowledge requires more than widespread exposure. Pinker distinguishes a self-evident public event from a focal point that merely creates common salience. His treatment of social media is equally careful: personalized distribution may feel public without duplicating a genuinely shared national broadcast.

Synthetic publicity extends that distinction. A case qualifies only when four conditions jointly hold:

1. **Provenance decoupling:** materially dependent signals share a generator, campaign, retrieval lineage, ranking process, or other common cause that is not apparent to the observer.
2. **Social representation:** the interface or format presents the signals as evidence of plural human testimony, endorsement, or attention.
3. **Perceived mutual visibility:** the observer infers that relevant others also encountered or recognized the displayed signal.
4. **Distinctive behavioral consequence:** the combined condition changes coordination, expression, compliance, or dissent beyond the effect explained by semantic content, repetition, and ordinary popularity cues.

Without the third condition, the case is false corroboration rather than synthetic publicity. Without an incremental behavioral consequence, the construct adds no explanatory value.

For measurement, separate three non-commensurable components:

- $V(s)$: observable visibility;
- $I(s)$: effective independence of the sources represented;
- $E(s)$: human acts represented by the signal.

Here, V is measured exposure or displayed reach; I is effective source independence; and E must be recorded separately as testimony, endorsement, amplification, or mere human presence. These variables do not share a scale and should not be collapsed into one inequality. **Synthetic publicity** exists when the four conditions above are satisfied and observers perceive public visibility or social corroboration unsupported by the effective independence and human acts represented.

The definition is deliberately broader than bots and narrower than online popularity. It includes generated accounts, copied or model-mediated language, automated summaries of what “people are saying,” engagement metrics inflated by coordinated processes, and interfaces that conceal common provenance. It excludes ordinary popularity when the displayed response accurately represents independent human endorsement.

Synthetic publicity also differs from classic astroturfing. Astroturfing hides sponsorship to simulate grassroots support. Synthetic publicity identifies the epistemic condition produced, whether or not deception was intended and whether the producer is political, commercial, institutional, or merely algorithmic. A recommender system can create the condition without inventing a single account if it makes selected reactions appear representative and mutually observed.

5. The coupled mechanism

The mechanism has four steps.

First, a user encounters a reaction or aggregate signal. Second, HBU assigns it weight as evidence about a proposition or norm. Third, visible multiplicity and apparent independence support an inference about what others believe. Fourth, perceived publicity supports an inference about what others know that others believe. The user then chooses an action partly in anticipation of others’ actions.

Let m be the number of displayed reactions. If signal errors have an average intraclass correlation $\bar{\rho}$, a first operational approximation of the effective number of independent sources is the standard clustered-sample design-effect approximation (Kish, 1965):

$$n_{eff} = \frac{m}{1 + (m-1)\bar{\rho}}.$$

When $\bar{\rho} = 0$, each reaction adds independent information and $n_{eff} = m$. As $\bar{\rho}$ approaches one, additional reactions add little independent evidence. The observer’s estimate, $\hat{n}_{eff,i}$, can be measured by asking how many independent judgments the display represents. Provenance concealment predicts $\hat{n}_{eff,i} > n_{eff}$. Other dependence structures may require clustering by generator, retrieval source, campaign, or repost tree rather than intraclass correlation.

The same error can propagate to higher-order beliefs. What looks like consensus may be a correlated sample; what looks like public recognition may be a collection of personalized impressions. The resulting action can still be rational relative to the perceived environment. If a worker believes that everyone else will comply with a new norm, dissent may be costly even if the norm has little genuine private support.

This account therefore distinguishes epistemic error from strategic adaptation. A person may correctly infer that a visible metric will influence others while knowing that the metric is manipulated. Synthetic publicity can coordinate behavior without persuading anyone of the underlying proposition.

6. Relations to adjacent concepts

The majority illusion shows how network structure can make a local attribute appear globally prevalent. Pluralistic ignorance concerns errors about others’ private attitudes. Preference falsification and the spiral of silence explain why public expression can diverge from private belief under reputational or isolation costs (Kuran, 1995; Noelle-Neumann, 1974). Persuasion-bias models show how people may fail to discount repeated information (DeMarzo, Vayanos, & Zwiebel, 2003). Research on public signals examines welfare when agents care about coordination as well as fundamentals (Morris & Shin, 2002). Work on social bots documents artificial amplification of low-credibility content and experimentally observed changes in perceptual bias and regulatory propensity (Shao et al., 2018; Yan et al., 2023), although automated bot classification has itself generated measurement disputes (Gallwitz & Kreil, 2022). These studies concern different outcomes and do not by themselves establish effects on perceived mutual visibility or higher-order coordination.

Pawełec and Łabuz (2026) provide a particularly close contemporary antecedent. Their analysis of political “softfakes” includes synthetic testimonials of support and generated representations of follower diversity. It therefore anticipates important parts of provenance decoupling and social representation. The present review could verify the article’s bibliographic record and abstract, but not independently inspect its full text. On that evidence, no claim is made that Pawełec and Łabuz omit mutual visibility or higher-order belief; rather, those elements remain unverified. Synthetic publicity can retain a residual role only if the conjunction of perceived mutual visibility and incremental behavioral consequence proves analytically and empirically useful.

Aumann’s agreement theorem is a negative benchmark here, not a model of platform behavior. It requires common priors and posteriors that are common knowledge. Synthetic-publicity environments provide no assurance of common priors, transparent information partitions, independent evidence, or common knowledge of posteriors. The theorem therefore clarifies why apparent online consensus cannot be treated as rational convergence under Aumann’s conditions.

Synthetic publicity does not replace these concepts. The literature audit conducted for this paper

classifies it as a refinement, not an unprecedented phenomenon. Astroturfing and computational propaganda already cover concealed coordination and simulated social endorsement. Manufactured-consensus research comes closest to the proposed label, while softfake research now directly addresses synthetic representations of political support. The majority illusion supplies a non-synthetic mechanism for false prevalence, while persuasion bias supplies a technical account of failure to discount repeated sources. The proposed term is retained provisionally because it designates an epistemic condition that need not involve a deliberate manufacturer and requires a measured inference of mutual visibility. It should be abandoned in favor of an existing category if that residual mechanism cannot be demonstrated. Synthetic publicity remains useful only if perceived mutual visibility and its incremental effect on higher-order belief or coordination are empirically demonstrated.

Adjacent family	What it already explains	Proposed residual
Astroturfing	Concealed sponsorship and simulated grassroots support	May arise without sponsorship or intentional deception
Computational propaganda	Automated, targeted and coordinated influence	Separates displayed plurality from perceived mutual visibility
Manufactured consensus	Artificial appearance of broad support	Requires a measured higher-order and coordination consequence
Political softfakes	Synthetic testimonials and representations of follower support or diversity	Tests hidden dependence together with perceived mutual visibility and incremental conduct
Coordinated inauthentic behavior	Deceptive coordination among accounts or assets	Can arise from interface selection without deceptive accounts or coordinated operators
Algorithmic amplification	Ranking and recommendation make selected material disproportionately visible	Requires that selection be read as plural social corroboration and mutually visible
Generative multi-user simulation	One system can produce plausible discourse from many apparent participants	Requires a measured inference about a real social audience and a behavioral consequence
Majority illusion	Locally distorted prevalence caused by network structure	Adds hidden common provenance and represented plurality
Persuasion bias	Failure to discount repeated information	Adds attributed autonomy, social representation and publicness
Pluralistic ignorance and preference falsification	Divergence between private belief and perceived or expressed opinion	Specifies a generative information condition that may produce or sustain the divergence

The contribution is therefore not a claim to have discovered fake consensus. It is a proposed integration that separates three questions:

1. Is the content true?
2. How many independent agents produced or endorsed it?
3. What does each observer believe other observers saw and inferred?

Existing content moderation frequently concentrates on the first question. Provenance systems address part of the second. Coordination depends heavily on the third.

7. Falsifiable predictions

The model yields five predictions.

P1. Higher-order displacement. Holding content constant, labeling a signal as widely and publicly observed will change estimates of others’ beliefs more than the observer’s own factual belief.

P2. Independence correction. Revealing that many outputs share a generator or source lineage will reduce perceived consensus more than it reduces evaluation of the semantic argument.

P3. Optimization disclosure. Revealing that reactions were selected or generated to maximize engagement, agreement, or retention will reduce their HBU weight, even when their wording is unchanged.

P4. Coordination threshold. Effects on action will be strongest when miscoordination is costly and when the displayed signal appears mutually observable.

P5. Provenance-publicity interaction. Provenance disclosure alone will be insufficient when interfaces continue to display dependent reactions as numerous and public. The largest correction will require joint disclosure of origin, dependence, and distribution.

8. Experimental design

A minimum experiment should hold semantic content constant and cross eight factors: visible quantity; human versus synthetic origin; independent versus shared provenance; concealed versus disclosed dependence; private versus mutually visible distribution; low versus high cost of miscoordination; sparse versus clustered network exposure; and a common feed versus personalized feeds. A fractional factorial pilot should be used before a fully crossed design to preserve power and identify the decisive provenance-by-mutual-visibility interaction. Critical comparison groups include independent human sources, coordinated human sources, multiple outputs from one generator, transparently labeled synthetic sources, and common-origin messages with and without visible popularity metrics. Participants would evaluate an identical substantive claim and then report:

- their own probability that the claim is true;
- their estimate of the average participant’s belief;
- their estimate of what participants believe others believe;
- perceived number of independent endorsers;
- willingness to speak, comply, coordinate, or dissent.

The key estimand is not merely attitude change. It is the difference between first-order and higher-order updating:

$$\Delta_{HO} - \Delta_{FO}.$$

A second experiment could hold the feed constant while varying disclosure that the visible accounts derive from one model, several independently trained models, or human participants. A third could disclose the recommendation objective. Preregistration should identify whether the theory predicts changes in factual belief, perceived consensus, coordination, or all three.

The primary analysis should use a preregistered multilevel model with first-order belief, higher-order belief, and action estimated separately. The decisive test is the interaction between common provenance and perceived mutual visibility after controlling for quantity, popularity, network exposure, semantic credibility, and participant-level heterogeneity. A mediation analysis may test whether perceived source independence and perceived mutual visibility transmit the effect to action, but it must report sensitivity to unmeasured mediator-outcome confounding and must not convert correlational mediation into a causal claim.

Manipulation checks must measure perceived origin, perceived independence, comprehension and credibility of the disclosure, estimated audience overlap, and perceived mutual visibility.

The construct adds explanatory value only if the joint provenance-decoupling and mutual-visibility manipulation explains incremental variance in higher-order belief or coordination after controlling for repetition, familiarity, perceived majority size, authority and semantic credibility. It should be rejected or absorbed into simpler theories if dependence disclosure does not change $\hat{n}_{eff,i}$, perceived mutual visibility does not mediate action, the predicted interaction is absent, or ordinary repetition and social proof reproduce the entire pattern without hidden dependence.

9. Governance implications

The immediate implication is not a ban on synthetic content. Generated speech can be useful, truthful, and socially beneficial. Nor should platforms be required to reveal identities where anonymity protects vulnerable speakers.

The narrower implication is that content provenance and social provenance are different. A label stating that an image was generated does not tell a user whether one process produced a thousand apparent endorsements. A useful disclosure regime would distinguish:

- origin: human, model, or mixed;
- lineage: independent or derived from a common process;
- representation: how many distinct humans actually endorsed or witnessed the claim;
- distribution: whether visibility is public, targeted, personalized, or paid;
- optimization: the objective that selected or generated the displayed reactions.

Any intervention should be tested for effectiveness and unintended consequences. Provenance labels may be ignored, may create false confidence in unlabeled material, or may expose legitimate anonymous speakers. The model recommends measurement before mandate.

10. Limitations

This is a conceptual paper. It presents no new behavioral data and establishes no prevalence estimate. The effective independence of sources is difficult to measure, particularly when humans edit model outputs or models share training data. Common knowledge in formal theory is stronger than the intuitive sense of publicity used in ordinary life. The paper therefore speaks of perceived publicity unless the conditions for strict common knowledge are demonstrated.

The concept may also prove too broad. Experiments must distinguish it from repetition, familiarity, authority, network homophily, ordinary social proof, and generalized distrust. Claims about social bots require special care because prominent detection methods have produced disputed classifications. Finally, synthetic publicity can sometimes reveal genuine latent agreement rather than fabricate it. Its welfare effect is not uniformly negative.

11. Conclusion

Generative systems do not need to change what everyone believes to change what everyone does. They can alter what each person thinks everyone else believes and whether that belief appears publicly recognized.

Common-knowledge theory explains why this matters for coordination. Heteronomous Bayesian Updating explains why reactions that look autonomous and independent acquire epistemic weight. Synthetic publicity names their conjunction: visibility and apparent corroboration detached from an equivalent population of independent human speakers or witnesses.

The proposal is modest enough to test and important enough to test carefully. The next task is not to announce that artificial intelligence has destroyed shared reality. It is to measure, separately, truth assessment, perceived independence, higher-order belief, and coordinated action.

AI-assistance disclosure

AI systems assisted with literature discovery, source checking, drafting, and editorial revision. The author directed the research questions, selected and reviewed the arguments, supplied the private source materials identified in the source audit, and accepts responsibility for the manuscript. AI-generated summaries were not treated as primary evidence. Bibliographic and substantive claims were checked against the sources and audit records identified in the accompanying research package.

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